

Predicting Popular Recipes

A data-driven approach to maximizing website traffic

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Business Goals

Displaying a popular recipe on the homepage can increase overall website traffic by up to 40%.

How to decide if a recipe will be popular?

We aim to:

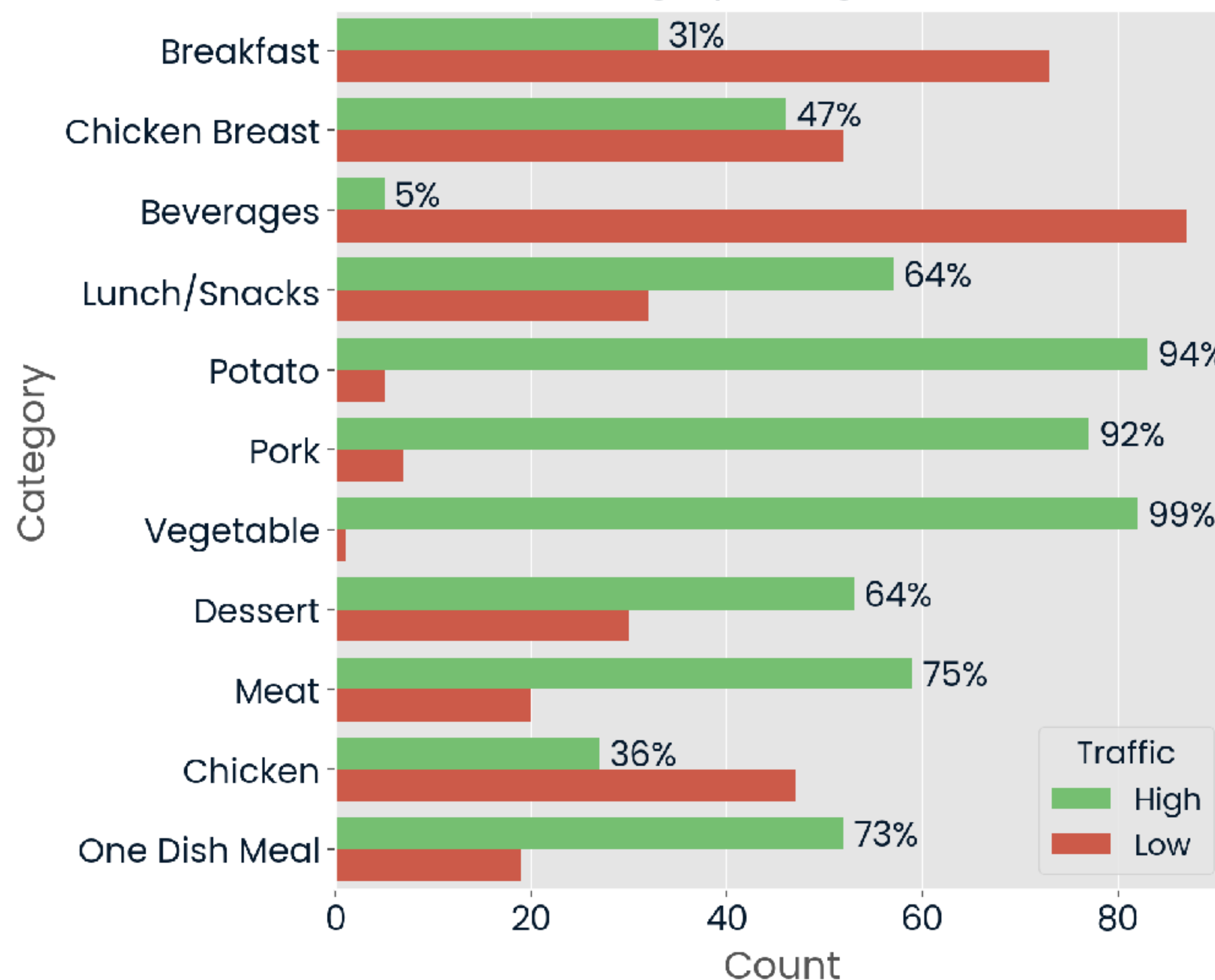
- Predict which recipes will drive high traffic
- Correctly predict high traffic recipes 80% of the time

The Data

recipe	calories	carbohydrate	sugar	protein	category	servings	high_traffic
1	399.26	19.56	5.23	29.82	Pork	6	High
2	35.48	38.56	0.66	0.92	Potato	4	High
3	914.28	42.68	3.09	2.88	Breakfast	1	Low
4	97.03	30.56	38.63	0.02	Beverages	4	High
5	27.05	1.85	0.8	0.53	Beverages	4	Low
6	691.15	3.46	1.65	53.93	One Dish Meal	2	High
7	183.94	47.95	9.75	46.71	Chicken Breast	4	Low
8	299.14	3.17	0.4	32.4	Lunch/Snacks	4	Low

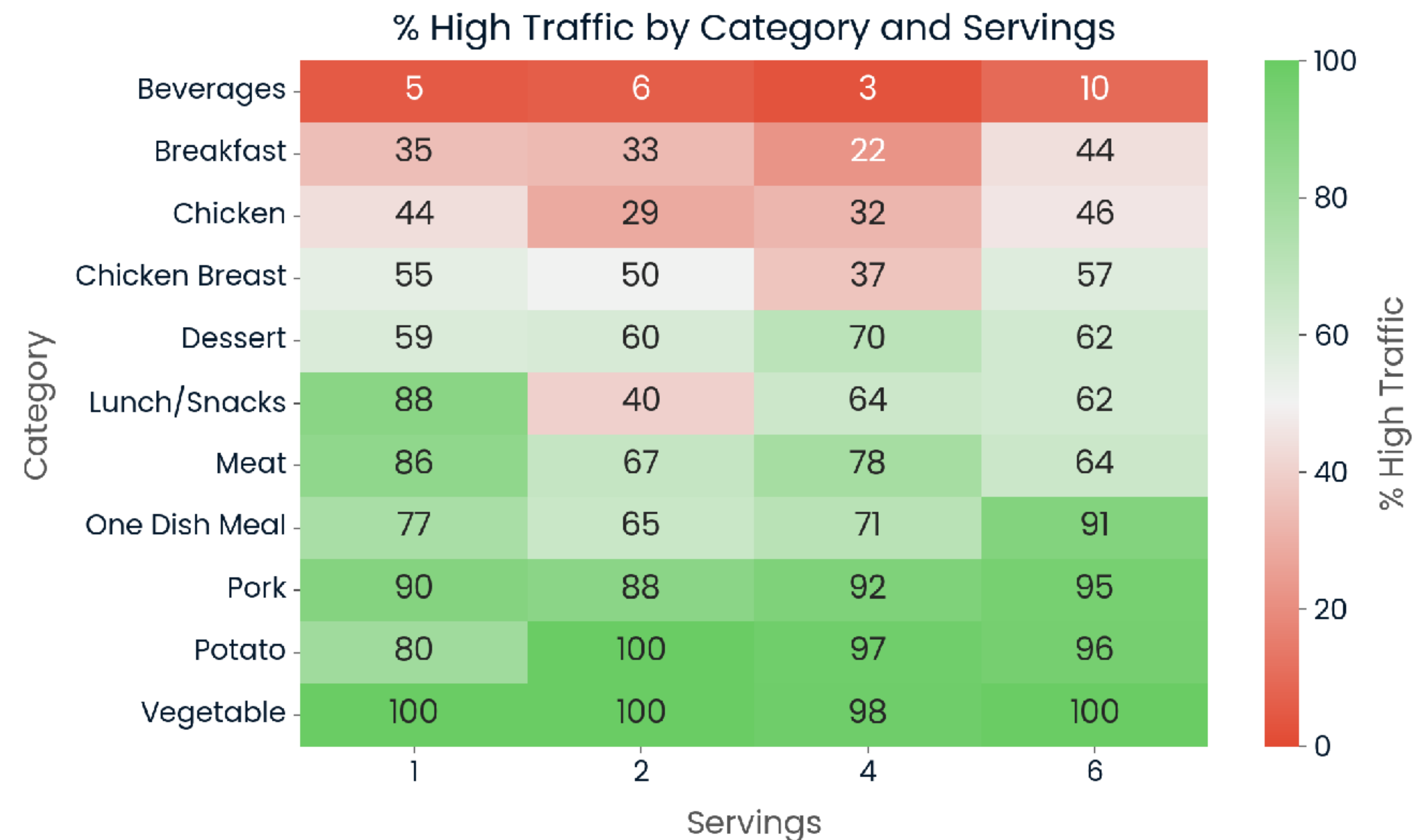
Data Insights

Category vs High Traffic



- Categories with most recipes: “*Breakfast*”, “*Chicken Breast*”, and “*Beverages*”.
- High traffic winners: “*Vegetable*” (99%), “*Potato*” (94%), and “*Pork*” (92%).
- Low-traffic outlier: “*Beverages*” - only 5% vs. 31% for the next lowest (“*Breakfast*” 31%, “*Chicken*” 36%).

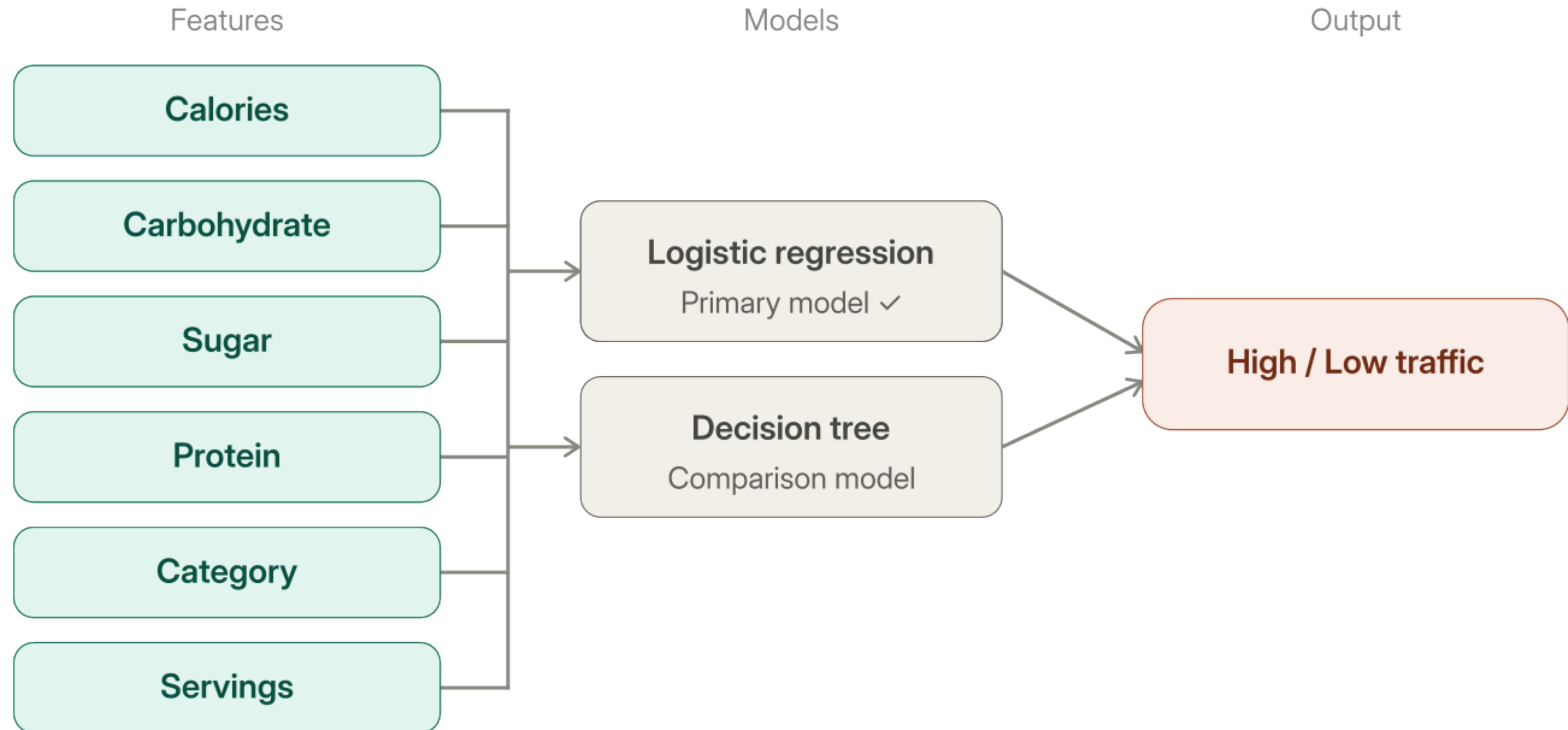
Data Insights



Category is a stronger driver of traffic:
certain categories dominate regardless of serving size

Model Overview

Two models: Logistic Regression and Decision Tree Classifier



Evaluation Metrics

How do we measure model performance?

Accuracy	*** Precision ***	Recall
<i>Of all recipes, how many were classified correctly?</i>	<i>Of recipes predicted as popular, how many actually were?</i>	<i>Of all truly popular recipes, how many did we correctly find?</i>

Key business metric

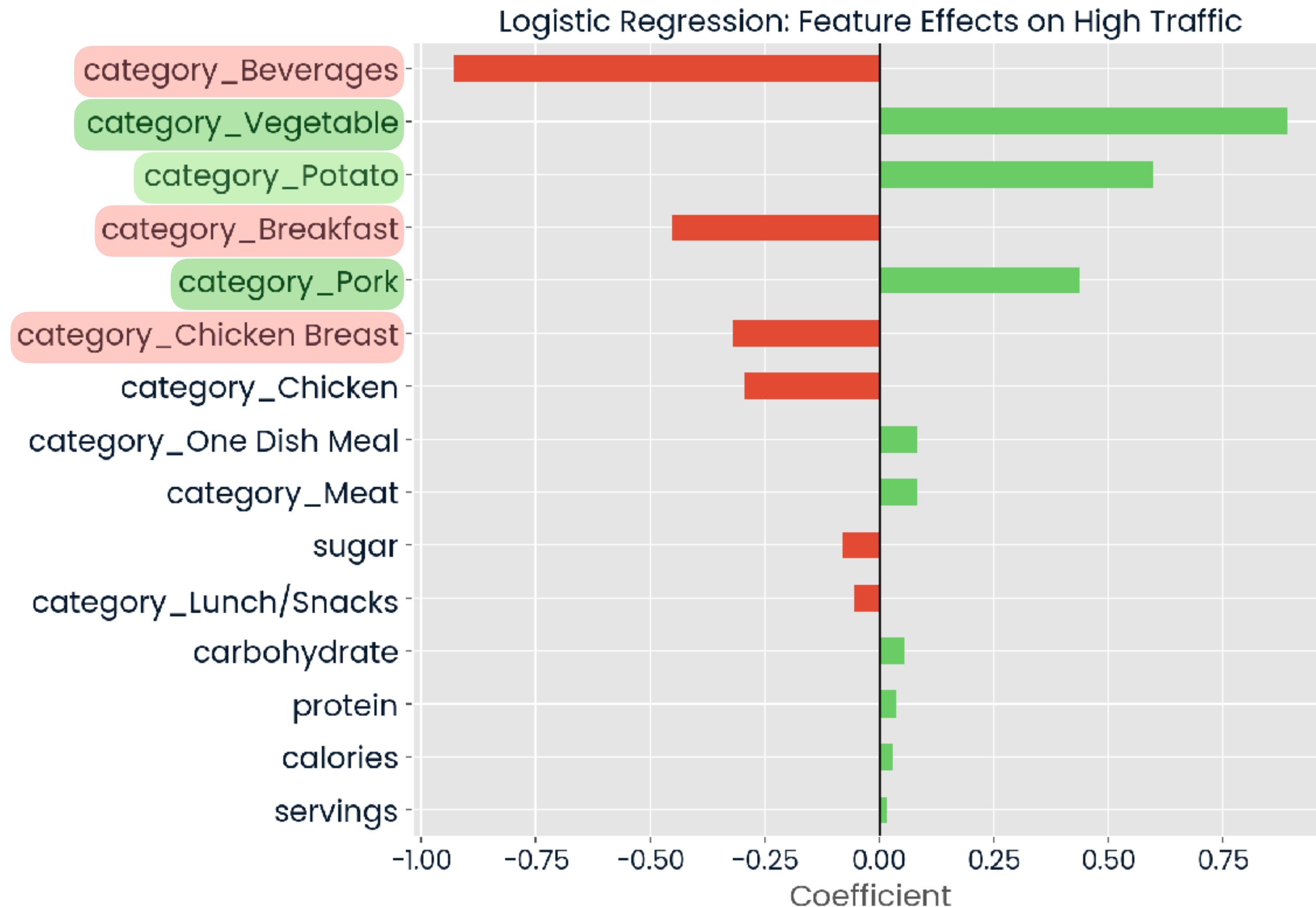
Results

Model	Accuracy	Precision	Recall
Logistic Regression (scaled)	0.779	0.848 	0.774
Decision Tree (max_depth=4)	0.784	0.849 	0.783

- Both models perform similarly on the test set
- Logistic Regression selected as the final model - simpler and highly interpretable

Feature Effects

Category - the key KPI driver



- Recipe category is the single most influential factor in traffic performance
- **Vegetable, Potato and Pork** drive the highest traffic; **Beverages** the lowest
- Nutritional content and serving size have minimal impact

Recommendations

 **Prioritize**

Vegetable	99% high traffic
Potato	94% high traffic
Pork	92% high traffic

 **Avoid**

Breakfast	31% high traffic
Beverages	5% high traffic
Chicken	36% high traffic

Content strategy

Source more Vegetable, Potato and Pork recipes - underrepresented but highest traffic drivers

Model screening

Screen recipes before publishing - 85% precision protects high-visibility slots and supports the 40% traffic uplift target

Looking ahead

Use predicted probability for ranking, not just yes/no
Retrain periodically as trends evolve